LECTURE 8

Control

for autonomous driving

How a vehicle follows a plan and corrects mistakes

CISC 667 · Introduction to Autonomous Driving

What students should understand

How feedback helps a vehicle recover from a mistake.

How different controllers decide to accelerate or steer.

Why delay, physical limits, and a poor model can spoil a good plan.

Planning chooses; control follows

Planning chooses a path and a desired speed.

Control keeps adjusting the vehicle toward that plan.

The road, the vehicle, and imperfect information affect the result.

Feedback: look, act, look again

1

Observe the vehicle

Where are we now?

→

2

Compare with the plan

What needs correcting?

→

3

Adjust the controls

Accelerate, brake, or steer

→

4

Observe again

Did the correction help?

A correction changes the situation. The next measurement closes the loop.

Reacting and anticipating work together

Approach	Driving intuition	Limitation
Fixed command	Keep the same pedal position	A hill changes the speed
Feedback	Notice the slowdown and correct it	The error has already appeared
Feedforward	Anticipate a known demand	The model can be wrong

A model is a useful simplification

1

Steering and speed

Possible control actions

→

2

Vehicle model

Predict the next movement

→

3

Predicted motion

Useful, but imperfect

A prediction helps choose an action; new measurements reveal what really happened.

What does good control look like?

Look at	Ask	Do not overlook
Following the plan	Does it stay close?	A brief large miss
Recovery	Does the error fade away?	Repeated side-to-side motion
Control action	Is steering reasonably gentle?	Large or rapidly changing requests
Physical response	Can the actuator keep up?	Delay and limited steering

How we will watch the demonstrations

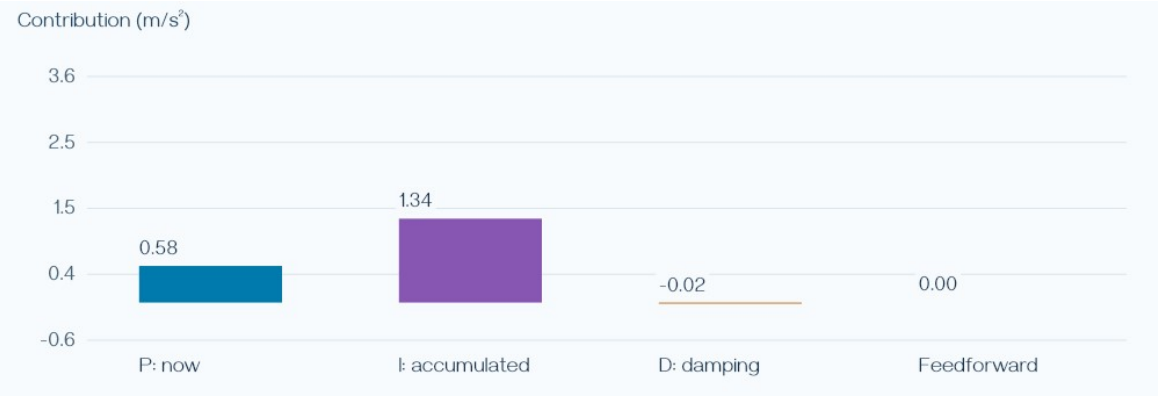
Choose a named case and press Play.

Watch one feature: speed, target point, steering, or prediction.

Switch to a contrasting case and explain what changed.

Open interactive demonstration

PID: three ways to correct speed



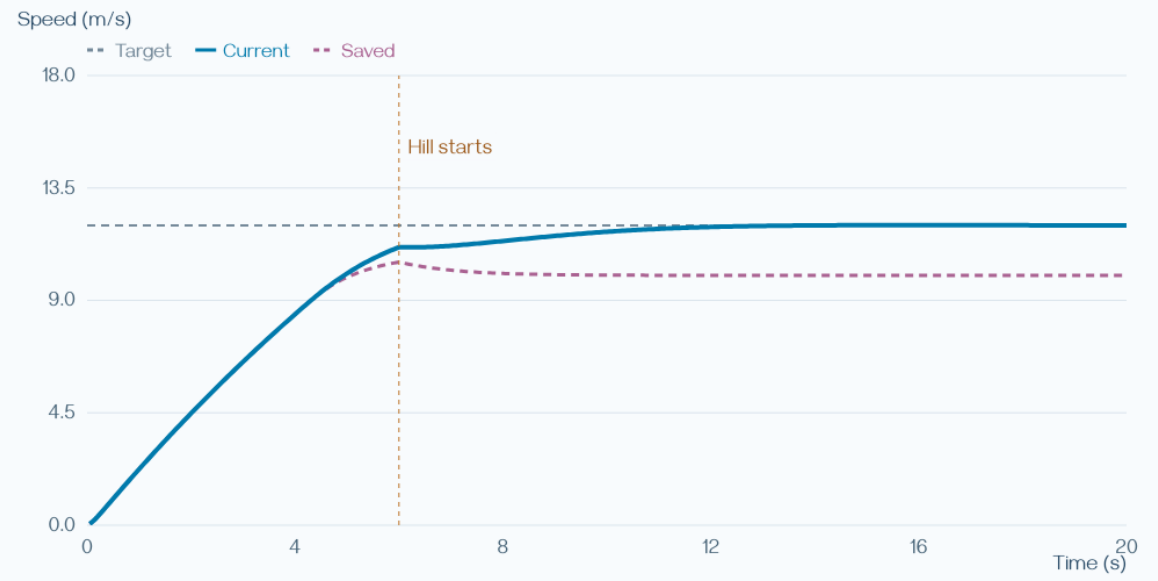
P reacts to how far speed is from the target.

I builds a correction for a persistent shortfall.

D damps rapid changes in measured speed.

Open interactive demonstration

Watch: holding speed on a hill



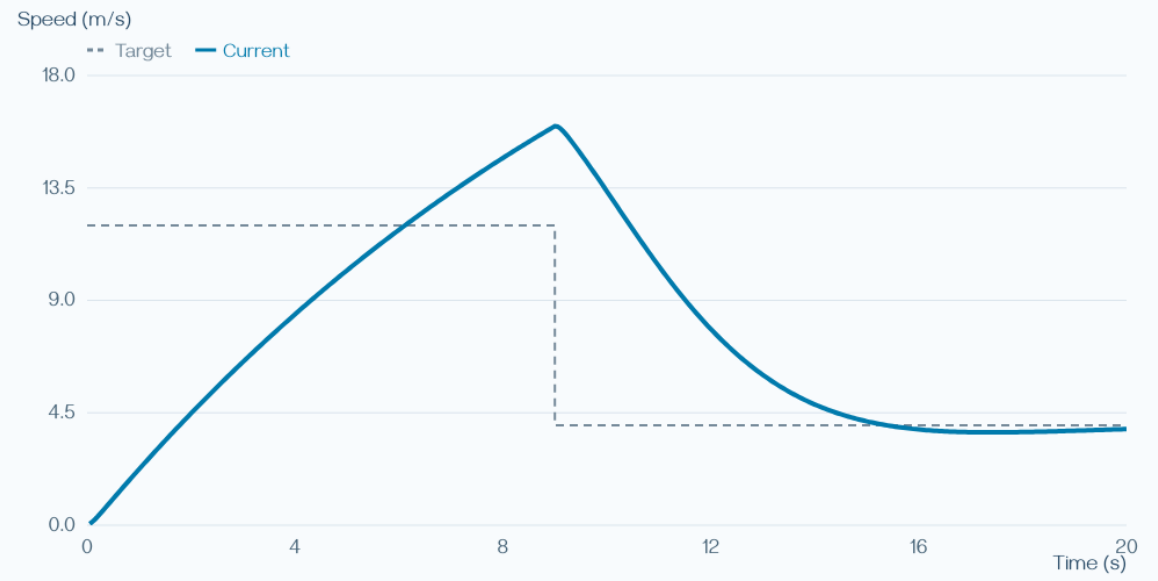
With P alone, speed settles below the target.

Adding I builds the extra push that is missing.

Watch the gap close after the hill begins.

Open interactive demonstration

When correction keeps building up

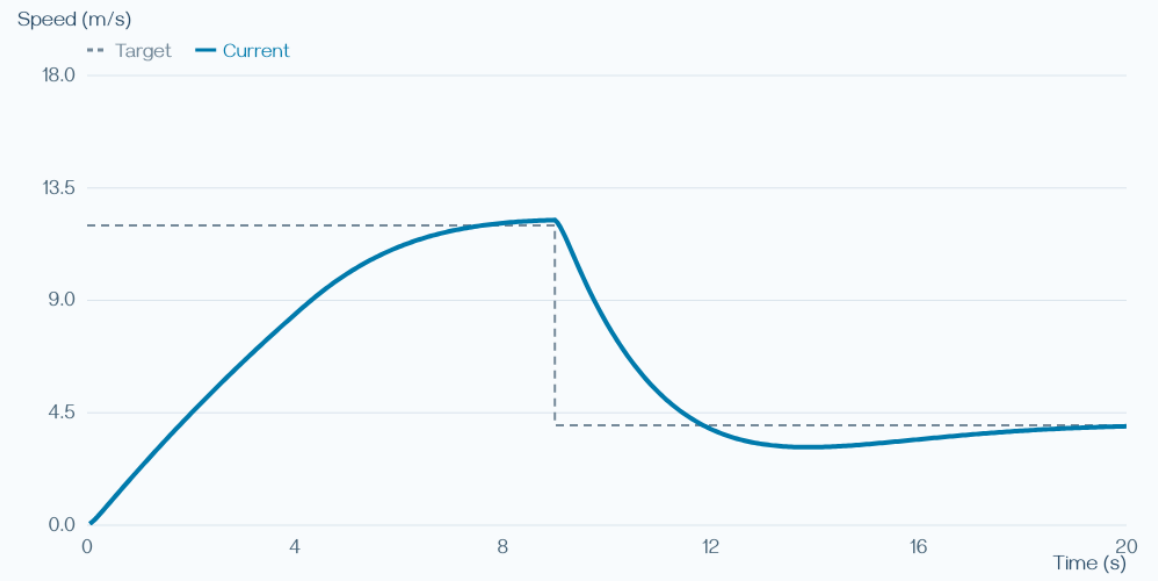


The actuator is already doing all it can.

The controller keeps remembering the shortfall.

That stored correction can delay recovery.

Anti-windup respects the actuator



Pause the buildup when it makes saturation worse.

Keep useful correction without storing as much excess.

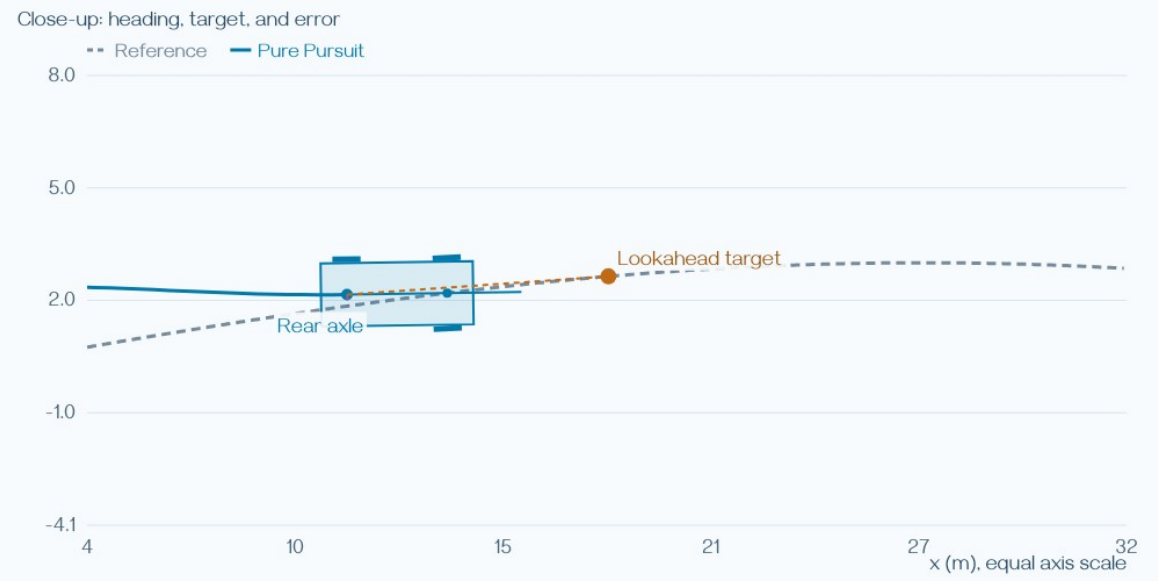
Braking still takes time.

Open interactive demonstration

Four ideas for following a path

Controller	Main idea	What to watch
Pure Pursuit	Aim at a point ahead	Where the target is
Stanley	Align and move toward the path	Heading and sideways error
LQR	Choose model-based feedback	Tracking versus steering effort
MPC	Plan ahead, then replan	Prediction and physical limits

Pure Pursuit: follow a point ahead



Pick a target farther along the path.

Turn toward that target.

Move forward and pick the next target.

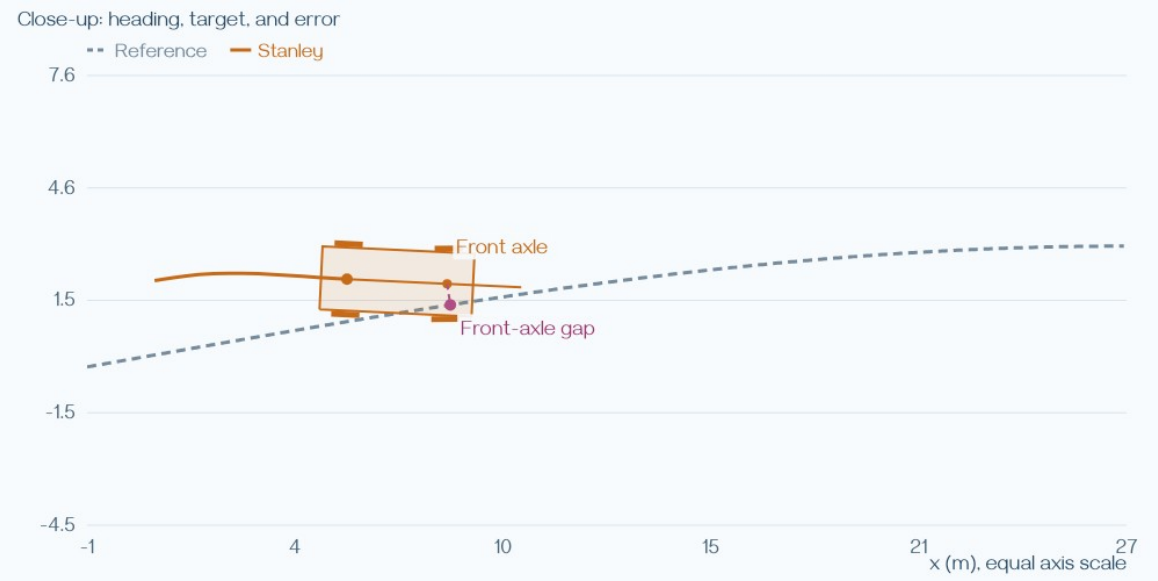
Open interactive demonstration

Watch: near target or far target?

Target choice	Typical behavior	Possible drawback
Nearby	Corrects more sharply	Busy or sensitive steering
Farther ahead	Turns more gently	May cut corners or recover slowly
Higher speed	Less time to correct	The same choice may work worse

Open interactive demonstration

Stanley: align and move toward the path



Is the vehicle pointing along the path?

Is the front axle to one side of it?

Combine those two corrections.

Open interactive demonstration

Watch: a strong correction can arrive late



A stronger response tries to remove the error quickly.

Delay means the vehicle has moved before it acts.

Repeated late corrections can create oscillation.

Open interactive demonstration

Using a model to choose a response

1

Current situation

Position and heading

→

2

A model of response

What a correction will do

→

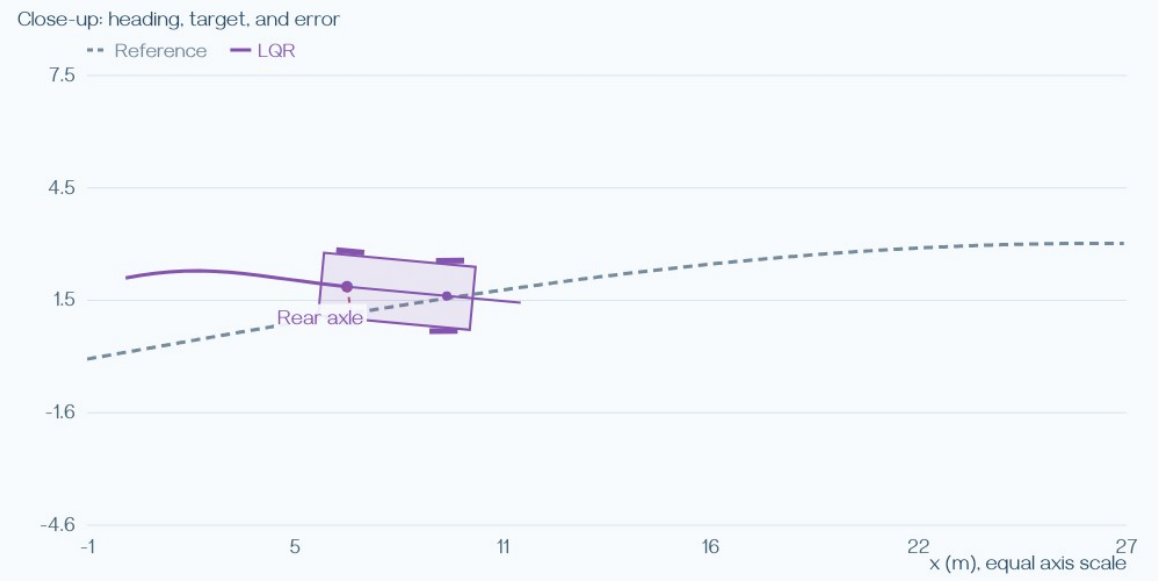
3

A preferred tradeoff

Close tracking or gentler steering

The model helps turn a design preference into a control rule.

LQR: balance tracking and steering effort



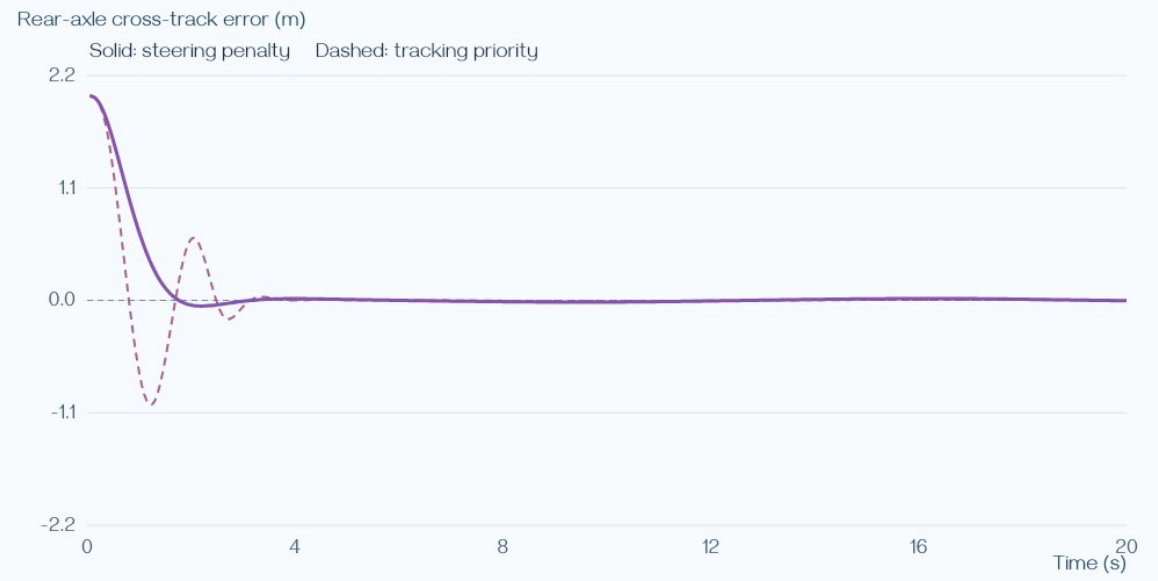
Use a model to choose the feedback response.

Decide how strongly to value accurate tracking.

Balance that against large steering corrections.

Open interactive demonstration

Watch: change the priority



Tracking priority tolerates stronger corrections.

Steering penalty discourages large corrections.

A steering limit can prevent the designed response.

Open interactive demonstration

Why look farther ahead?

A bend is coming while steering is still changing.

Waiting until the error appears can be too late.

Prediction lets the controller consider future choices now.

MPC: plan, act once, update

1

Predict ahead

Use the current state

→

2

Choose a sequence

Consider preferences and limits

→

3

Apply the first action

Start carrying out the plan

→

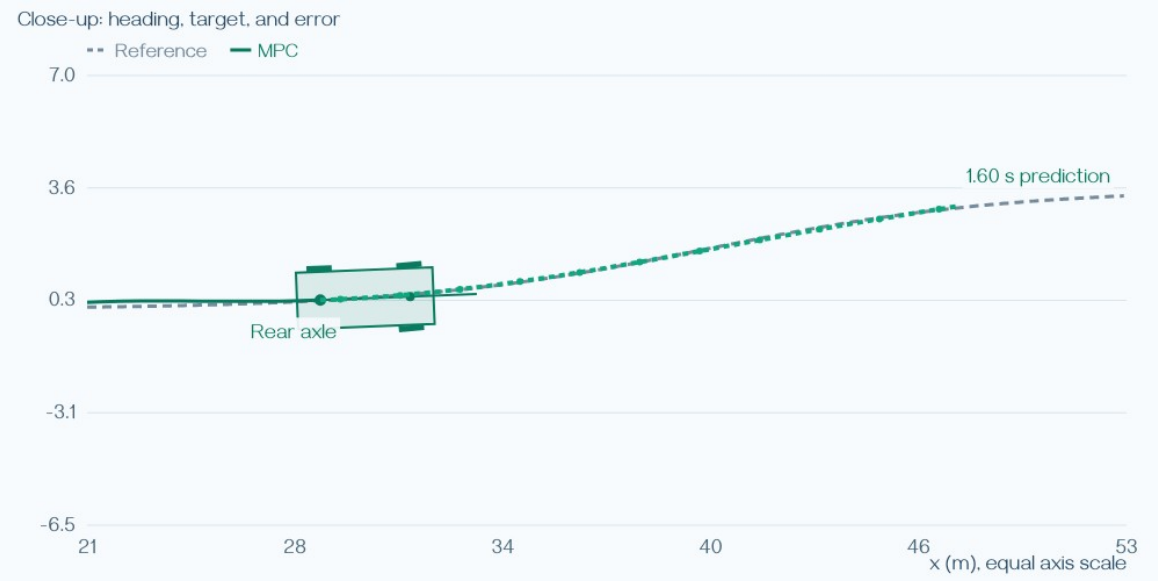
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Measure and replan

Use the latest situation

Future actions are a plan. The next measurement can change that plan.

The green path is a prediction



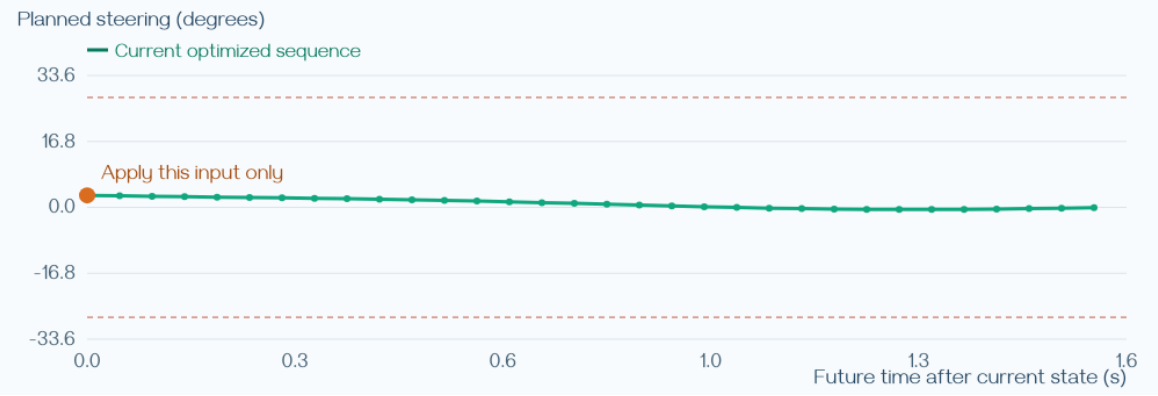
It shows where the current plan expects to go.

Longer preview considers more of the upcoming path.

It changes when the vehicle state changes.

Open interactive demonstration

Only the first planned action is used



The orange point marks the next steering request.

Later points are provisional choices.

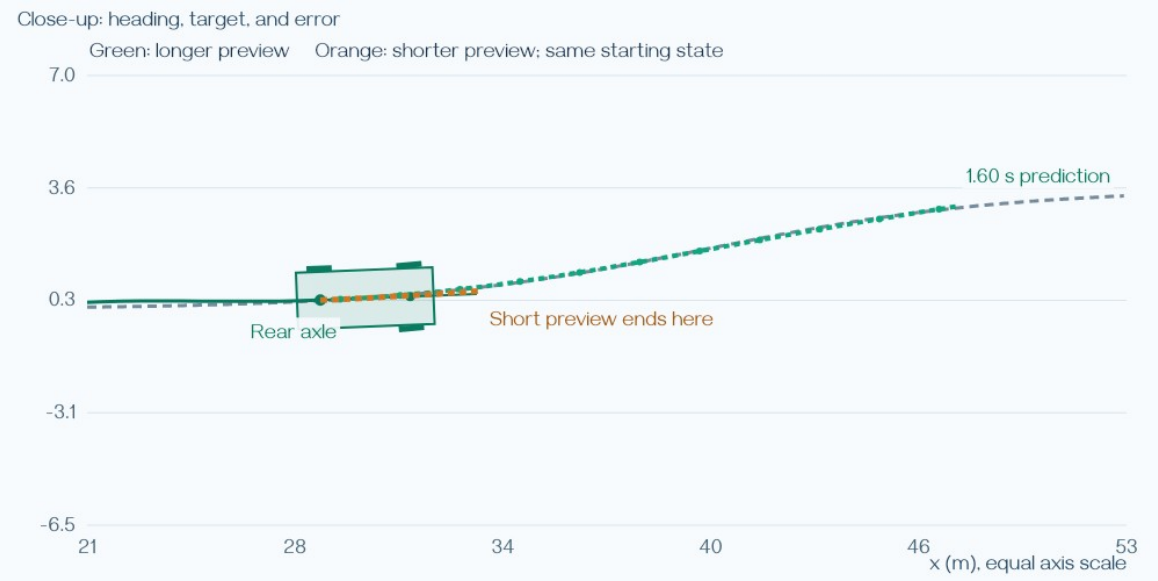
Measure again before trusting the rest.

Open interactive demonstration

A preference is different from a limit

Design choice	Meaning	Example
Tracking preference	Try harder to follow the path	Accept stronger corrections
Steering preference	Discourage large commands	Accept some tracking error
Physical limit	Stay within allowed inputs	Limited angle or turning rate
Preview length	Consider more of what is ahead	See farther into a lane change

Watch: short and long preview



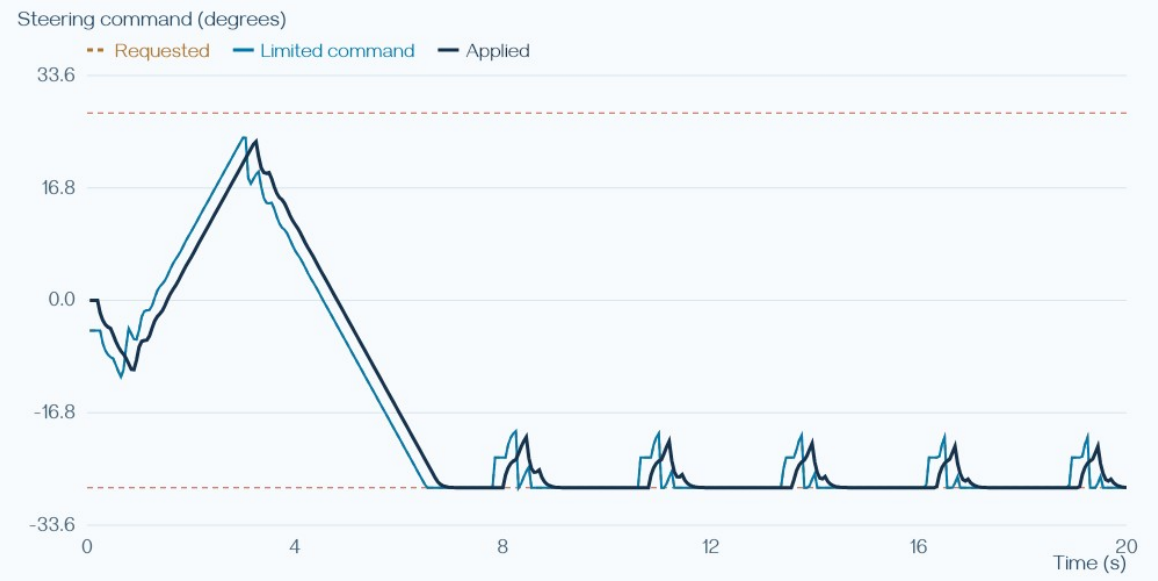
Compare how far the green prediction reaches.

Watch when steering begins and how the path is followed.

Tighten the limits and see what changes.

Open interactive demonstration

Prediction and execution can disagree



The controller requests a steering action.

The actuator needs time to carry it out.

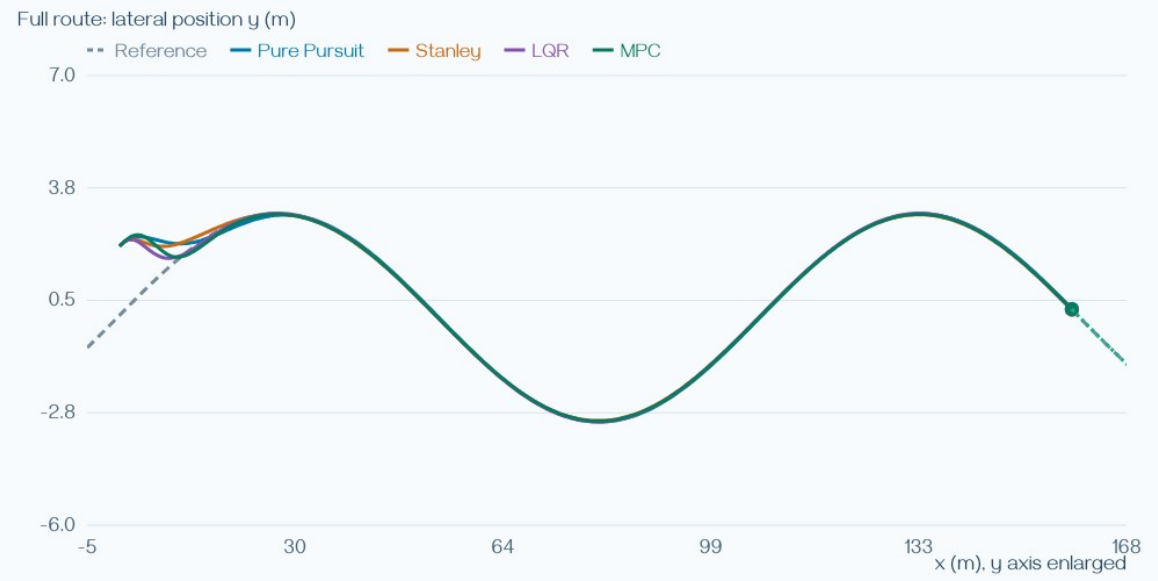
Delay or a biased response changes the actual motion.

Open interactive demonstration

Why a predictive controller can struggle

Problem	What can happen	Possible response
Poor prediction	Actual motion misses the plan	Improve the model or estimate
Impossible request	Tracking remains poor	Slow down or replan
Slow computation	A new command arrives late	Bound the work and use a fallback
Missing constraints	The plan ignores a real limit	Represent and test that limit

Compare under the same conditions



Each colored vehicle follows the same path.

All receive the same speed and actuator conditions.

Change one condition and watch the comparison change.

Open interactive demonstration

Read the shape of the response



A brief startup correction can settle down.

Repeated swings suggest continuing trouble.

A small typical error can still hide a large miss.

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Choose for the situation

Need	Useful starting point	What still matters
Hold a speed	PI/PID with anti-windup	Limits and disturbance response
Follow a simple path	Pure Pursuit or Stanley	Speed, geometry, and delay
Use a reliable local model	LQR	Tracking/steering tradeoff
Plan under input limits	MPC	Prediction quality and compute

Other ways to improve control

Approach	Core idea	Question to ask
Gain scheduling	Change response with conditions	Does it transition smoothly?
Adaptive control	Adjust as the vehicle changes	Can it learn the right change?
Robust control	Allow for modeled uncertainty	What uncertainty is covered?
Learning-based control	Learn a policy or a model	What happens in new situations?

Control depends on the whole vehicle stack

A fresh, trustworthy state estimate.

A feasible path and speed request.

An actuator that responds as expected.

Explain the idea in your own words

Why can remembering a speed shortfall help—and sometimes hurt?

How does aiming farther ahead change steering behavior?

Why does MPC keep changing a plan it just made?

Explore again after class

Extract Lecture8_Control.zip and open START_HERE.html.

Replay the named cases and compare what you see.

Use the concept guide or open optional details when curious.

Open interactive demonstration